

PP_0375 (Q88QV9): A Family-S9 Serine Peptidase of the *Pseudomonas putida* KT2440 Pyrroloquinoline-Quinone (PQQ) Biosynthesis Operon

Target: PP_0375 · **UniProt:** Q88QV9 · **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), taxid 160488 **Protein length:** 612 aa · **Gene synonyms:** *pqqG* (operon terminal gene); ortholog of *P. aeruginosa pqqH* (PA1990)

Summary

PP_0375 encodes a **family-S9 (prolyl oligopeptidase clan SC) serine peptidase** — a catalytically competent α/β -hydrolase whose primary molecular function is **peptide-bond hydrolysis**. The gene sits as the **terminal ("G") gene of the pyrroloquinoline-quinone (PQQ) biosynthesis operon** of *P. putida* KT2440, whose gene order has been mapped by RT-PCR as *pqqFABCDEG* (loci PP_0375–PP_0381). PP_0375 is the ~53 %-identical ortholog of *P. aeruginosa pqqH/PA1990*, the peptidase gene cotranscribed with *pqqABCDE*. Its role therefore lies within **PQQ cofactor maturation**, a ribosomally-synthesized-and-post-translationally-modified-peptide (RiPP) pathway that builds the redox cofactor PQQ from a glutamate and a tyrosine embedded in the 23-residue precursor peptide PqqA.

Structurally, PP_0375 has the **canonical two-domain prolyl-oligopeptidase architecture**: an N-terminal seven-bladed β -propeller domain that gates the active site, packed against a C-terminal α/β -hydrolase catalytic domain carrying the Peptidase_S9 module (Pfam PF00326). The catalytic **Ser469–His582–Asp550 triad** is geometrically intact in the high-confidence AlphaFold model (global pLDDT 93.8; triad distances 3.0 Å and 2.78 Å), with Ser469 sitting in a classic G-x-S-x-G "nucleophile elbow." Because the β -propeller covers the active site, this class of enzyme is intrinsically **selective for short peptides (oligopeptidase behavior)** rather than folded proteins — the enzyme sees a buried, gated pocket rather than an open surface groove.

Functionally, the PQQ operon of *P. putida* carries **two proteases of different mechanistic clans, doing two different jobs**. The essential core excision of the PqqE-cross-linked Glu–Tyr di-amino-acid from PqqA is performed by **PqqF** (PP_0381), an M16 zinc "inverzincin" whose deletion abolishes PQQ synthesis. In contrast, deletion of the S9 peptidase (*pqqH*, the PP_0375 ortholog) in *P. aeruginosa* does **not** block PQQ synthesis but abolishes **PQQ excretion** into the culture supernatant. PP_0375 therefore functions in the **cytoplasm** during cofactor maturation/release, and the PQQ it helps produce is exported to the **periplasm**, where it serves as the redox cofactor of quinoprotein dehydrogenases (glucose dehydrogenase Gcd, and the alcohol dehydrogenases PedE/PedH), driving glucose→gluconate oxidation (mineral-phosphate solubilisation) and alcohol/aldehyde oxidation. The **exact peptide substrate of PP_0375 has not been experimentally established** and remains the principal knowledge gap.

Gene/Protein Identity Verification

Before presenting findings, the mandatory identity checks were satisfied:

Check	Result
Gene symbol PP_0375 matches protein description	□ "Prolyl oligopeptidase family protein" is consistent with the S9 peptidase domains found (PF00326, IPR001375)
Organism correct	□ <i>P. putida</i> KT2440 (taxid 160488), OrderedLocusName PP_0375
Domains align with literature	□ AB_hydrolase_fold (IPR029058), Peptidase_S9_cat (IPR001375), Peptidase_S9 (PF00326) all confirmed; genomic context is the PQQ operon
Literature consistency	□ No mistaken-identity conflict — PP_0375 maps cleanly onto the <i>pqqG/pqqH</i> operon position; literature on the PQQ pathway is directly relevant

Conclusion: The identity is secure. PP_0375 is the operonic S9 peptidase (*pqqG*) of the *P. putida* KT2440 PQQ cluster.

Key Findings

Finding 1 — PP_0375 is a Peptidase S9 serine hydrolase with an intact Ser-His-Asp catalytic triad

UniProt Q88QV9 (612 aa) carries a consistent, mutually corroborating stack of domain annotations that place it firmly in the **prolyl oligopeptidase family (clan SC, family S9)**: Pfam **PF00326 (Peptidase_S9)**, InterPro **IPR001375 (Peptidase_S9_cat)**, **IPR029058 (α/β -hydrolase fold)**, IPR050585, PANTHER **PTHR43056:SF5**, eggNOG **COG1506** (dipeptidyl-/acylamino-acid peptidase), and Gene3D **3.40.50.1820** (the α/β -hydrolase superfamily fold).

The catalytic machinery is not merely annotated but **structurally verified**. The nucleophile **Ser469** lies in a canonical **G-x-S-x-G elbow** (sequence ...RGGSAGG...), the signature "nucleophile elbow" of α/β -hydrolases. In the AlphaFold model **AF-Q88QV9** (global pLDDT 93.8, i.e. very high confidence), the triad is geometrically complete and correctly oriented for catalysis:

- **Ser469 O γ – His582 N ϵ 2 = 3.0 Å** (hydrogen-bond distance for the charge-relay proton transfer)
- **His582 N δ 1 – Asp550 O δ 2 = 2.78 Å** (the second charge-relay contact)
- All three triad residues have local pLDDT > 86 (individually reliable)

The InterPro-derived Gene Ontology terms are **serine-type peptidase activity (GO:0008236)** and **proteolysis (GO:0006508)**. Together these establish, with high confidence, that PP_0375 is a functional serine peptidase and not a degenerate/pseudo-enzyme.

Finding 2 — PP_0375 is the *P. putida* ortholog of *pqqH*, a peptidase embedded in the PQQ biosynthesis operon

Genomic context is the decisive evidence. In the STRING network (organism 160488), PP_0375's highest-confidence functional partners are the **PQQ-biosynthesis genes**, and the supporting evidence channel is **genomic neighborhood (nscore)** — meaning these genes are co-located and co-inherited:

Partner	Locus	STRING score
pqqC	PP_0378	0.892
pqqE	PP_0376	0.871
pqqD	PP_0377	0.841
pqqB	PP_0379	0.795
pqqF	PP_0381	0.507

PP_0375 sits **immediately adjacent to *pqqE* (PP_0376)** — precisely the operonic position occupied by *pqqH* in *P. aeruginosa*. A Needleman–Wunsch global alignment shows PP_0375 (612 aa) is **52.9 % identical** (344 identities over 650 aligned columns) to *P. aeruginosa* **pqqH/PA1990** (Q9I2B9, 608 aa). This level of identity across nearly the full length indicates clear orthology, not merely shared domain content.

The functional identity of *pqqH* was established experimentally in *P. aeruginosa* ([PMID: 19902179](#)):

"Gene PA1990 of *Pseudomonas aeruginosa*, located downstream of *pqqE* and encoding a putative peptidase, was shown to be involved in excretion of PQQ into the culture supernatant. This gene is cotranscribed with the *pqqABCDE* cluster and was named *pqqH*."

PP_0375 is the *P. putida* ortholog occupying the same operonic position.

Finding 3 — PP_0375 (*pqqG/pqqH*) functions in PQQ cofactor biosynthesis for periplasmic quinoprotein glucose dehydrogenase

RT-PCR mapping of the *P. putida* KT2440 cluster defines its gene order as **pqqFABCDEG** (An & Moe 2016, [PMID: 27287323](#)). PP_0375 is the terminal "**G**" gene immediately after *pqqE* (PP_0376) — i.e. the peptidase-encoding gene, ortholog of *P. aeruginosa* *pqqH*/PA1990.

The biological purpose of the operon is to synthesize **PQQ**, the redox coenzyme of the periplasmic **glucose dehydrogenase (Gcd)** that oxidizes glucose to gluconic acid, enabling **mineral-phosphate solubilization** in the rhizosphere:

"Soil-dwelling microbes solubilize mineral phosphates by secreting gluconic acid, which is produced from glucose by a periplasmic glucose dehydrogenase (GDH) that requires pyrroloquinoline quinone (PQQ) as a redox coenzyme." — [PMID: 27287323](#)

"The *pqq* gene cluster (*pqqFABCDEG*) encodes at least two independent transcripts" — [PMID: 27287323](#)

Biochemically, PQQ is a **RiPP** built from a Glu and a Tyr embedded in the ribosomal precursor peptide **PqqA**. The precursor is cross-linked (C–C bond between the Glu C γ and Tyr C ϵ) by the radical-SAM enzyme **PqqE** (assisted by the RRE chaperone **PqqD**), after which a **protease/peptidase must excise the modified di-amino-acid** before **PqqB** (dual hydroxylase) and **PqqC** (eight-electron oxidase/cyclase) complete the cofactor:

"The PqqA peptide is recognised by PqqE, which links the C9 and C9a, afterwards it is accepted by PqqF which cuts out the linked amino acids." — [PMID: 18371220](#)

Finding 4 — Two operon proteases, two roles: PqqF (M16) excises PqqA; PP_0375 (*pqqG/pqqH*, S9) is linked to PQQ excretion

Neighbor annotations (UniProt, organism 160488) confirm the complete operon and, critically, that it encodes **two peptidases of entirely different catalytic clans**:

Locus	Gene	Length	Role
PP_0380	pqqA	23 aa	Precursor peptide (Glu + Tyr donor)
PP_0379	pqqB	—	Dual hydroxylase / metallo- β -lactamase-fold
PP_0378	pqqC	—	Ring cyclization/oxidation (8 e ⁻ , 8 H ⁺)
PP_0377	pqqD	—	PqqA-binding RRE chaperone
PP_0376	pqqE	—	Radical-SAM peptide cyclase (SPASM domain)
PP_0381	pqqF	766 aa	M16 zinc "inverzincin" protease (IDE-like)
PP_0375	pqqG	612 aa	S9 serine peptidase (this target)

Crystallography shows **PqqF has a clamshell inverzincin fold** with a large (~9,400 Å³) internal chamber proposed to bind PqqA and excise the PqqE-cross-linked Glu–Tyr; *pqqF* deletion abolishes PQQ biosynthesis (Wei et al. 2016):

"After linkage of the Cy of glutamate and Ce of tyrosine by PqqE, these two residues are hypothesized to be cleaved from PqqA by PqqF." — [PMID: 27231346](#)

"we demonstrated that the *pqqF* gene is essential for PQQ biosynthesis" — [PMID: 27231346](#)

By contrast, deletion of the **S9 peptidase *pqqH*** in *P. aeruginosa* does **not** block PQQ synthesis but abolishes **PQQ excretion** into the supernatant ([PMID: 19902179](#)). This establishes a **division of labor**: PqqF performs the essential core excision step, while PP_0375/pqqG/pqqH plays a distinct, non-essential-for-synthesis role tied to release/excretion of mature PQQ. **The precise molecular (peptide) substrate of PP_0375 has not been experimentally determined.**

Finding 5 — Two-domain prolyl-oligopeptidase architecture implies short-peptide (oligopeptidase) selectivity

InterPro/SUPFAM/Gene3D domain mapping of Q88QV9 reveals the classic two-domain organization of the prolyl-oligopeptidase family:

- **N-terminal β-propeller** ("DPP6 N-terminal domain-like," SCOP **SSF82171**, residues ~44–337)
- **C-terminal α/β-hydrolase catalytic domain** (Gene3D **G3DSA:3.40.50.1820**, res ~356–612; SUPFAM **SSF53474**, res ~358–603), carrying the Peptidase_S9 module (PF00326 / IPR001375, res ~401–607) with the **Ser469–Asp550–His582 triad**
- **PANTHER PTHR43056** (Xaa-Pro dipeptidyl-peptidase / cocaine esterase) spans res 4–608

In the AlphaFold model (pLDDT 93.8) the two domains **pack tightly** (75 Ca–Ca inter-domain contacts < 8 Å; centroid separation 32.5 Å), and the catalytic Ser469 lies in an **enclosed pocket** (165 protein heavy atoms within 10 Å of Ser469 Oy). This is a **covered/gated active site**, not an open surface groove.

The functional implication is well established for this fold family: the β-propeller acts as a **molecular sieve/gate** that admits only small, unstructured peptides to the buried catalytic center, excluding large folded proteins. PP_0375 is therefore predicted to be an **oligopeptidase** — **selective for short peptides**, which is mechanistically consistent with acting on the small PqqA-derived/PQQ-maturation intermediates rather than degrading bulk protein.

Finding 6 — The PQQ cofactor served by PP_0375 feeds multiple periplasmic quinoprotein dehydrogenases

UniProt (organism 160488) confirms **three PQQ-dependent apo-dehydrogenases** in KT2440 that consume the cofactor whose biosynthesis PP_0375 supports:

Enzyme	Locus	Length	Type / substrate
Glucose dehydrogenase Gcd	PP_1444	803 aa	Glucose → gluconate (periplasm)
Alcohol dehydrogenase PedE	PP_2674	631 aa	Ca/Zn-type PQQ-ADH, alcohols/aldehydes
Alcohol dehydrogenase PedH	PP_2679	595 aa	Lanthanide-dependent PQQ-ADH

Wehrmann et al. 2017 ([PMID: 28655819](#)) purified and characterized PedE/PedH as functionally redundant periplasmic PQQ-ADHs for oxidation of alcohols and aldehydes (relevant to volatile-organic-compound detoxification and catabolism); PedH was the first lanthanide-dependent PQQ-ADH described in a non-methylophile:

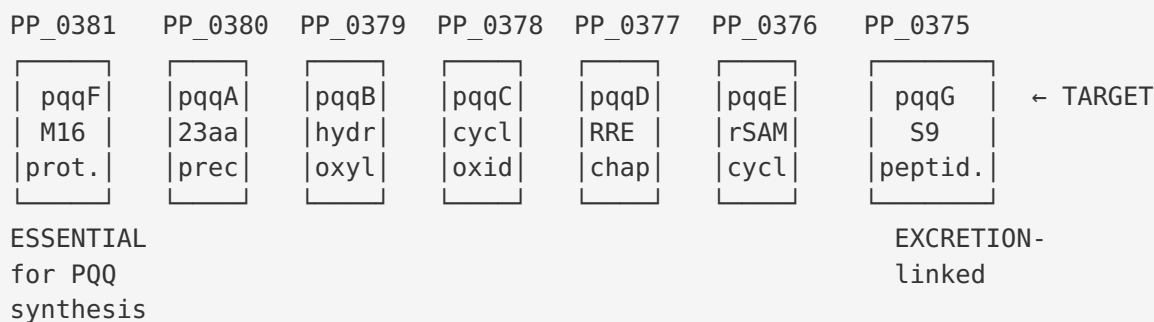
"many Gram-negative bacteria have evolved periplasmic oxidation systems based on pyrroloquinoline quinone-dependent alcohol dehydrogenases (PQQ-ADHs) that are often functionally redundant" — [PMID: 28655819](#)

This closes the loop: the cofactor PP_0375 helps mature is exported and used by periplasmic quinoproteins that drive both **glucose→gluconate oxidation** (Gcd; phosphate solubilisation, environmental acidification) and **alcohol/aldehyde oxidation** (PedE/PedH).

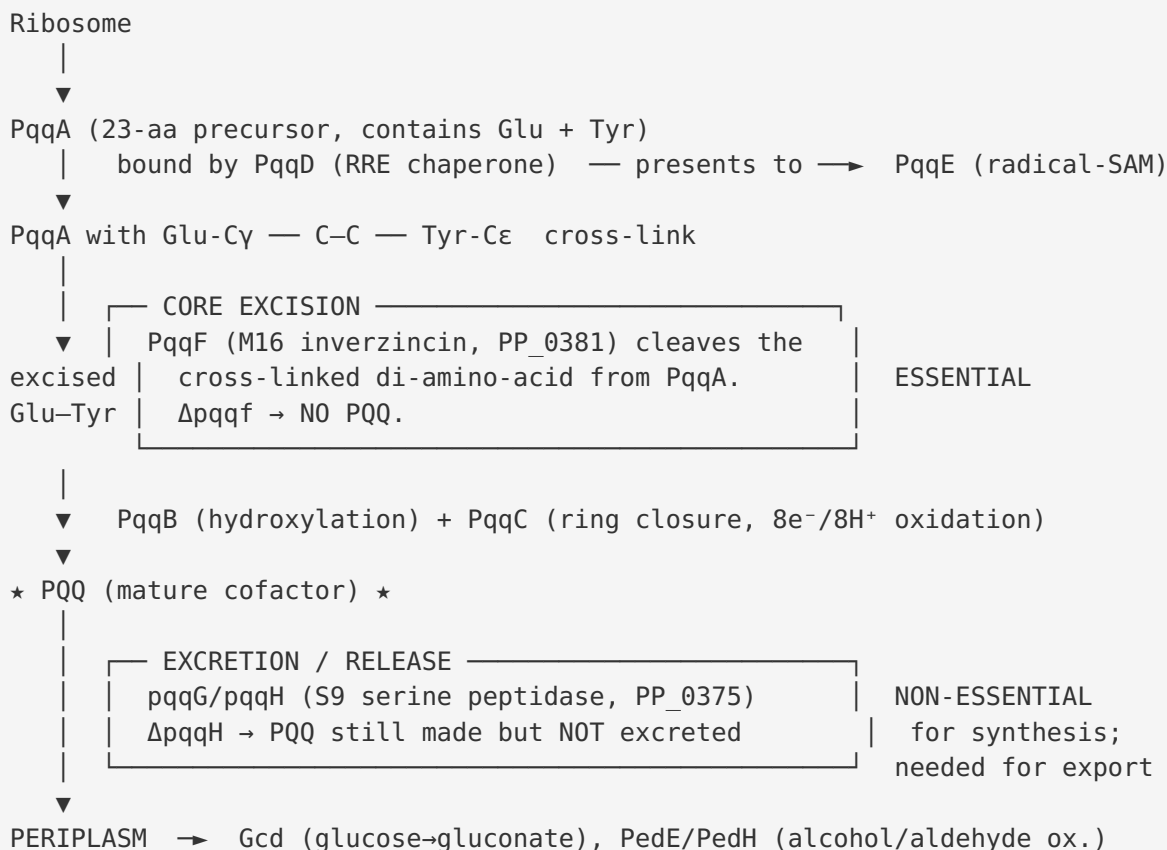
Mechanistic Model / Interpretation

The PQQ biosynthesis operon and PP_0375's place in it

P. putida KT2440 pqq operon (order: pqqFABCDEG)



Biosynthetic route (RiPP pathway) and the two-protease division of labor



Synthesis

PP_0375 is best understood as a **cytoplasmic, short-peptide-selective serine oligopeptidase acting in PQQ cofactor maturation**, distinct in both clan (S9 serine vs. M16 zinc) and phenotype (excretion vs. essential synthesis) from its operon partner PqqF. The structural evidence (intact triad, gated β -propeller active site, oligopeptidase architecture) is fully consistent with a role hydrolyzing a small peptide substrate during the late steps of PQQ maturation or release. The strongest experimental anchor for its physiological role is the *P. aeruginosa* *pqqH* knockout phenotype — **loss of PQQ excretion** — transferred by ~53 % orthology to PP_0375. Its exact substrate (whether a residual PqqA-derived peptide, a maturation intermediate, or a distinct target coupled to export) remains undetermined.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
19902179	<i>PQQ biosynthetic operons & transcriptional regulation in P. aeruginosa</i>	Defines <i>pqqH</i> (PA1990) as the peptidase downstream of <i>pqqE</i> , cotranscribed with <i>pqqABCDE</i> , required for PQQ excretion — the direct functional model for PP_0375
27287323	<i>Regulation of PQQ-dependent glucose dehydrogenase in P. putida KT2440</i> (An & Moe 2016)	Maps the KT2440 operon as <i>pqqFABCDEG</i> (placing PP_0375 as "G"); links PQQ to periplasmic Gcd and gluconate/phosphate solubilisation
18371220	<i>PQQ biosynthesis pathway revisited: a structural approach</i>	Describes the proteolytic excision step on the PqqE-cross-linked PqqA precursor
27231346	<i>Crystal structure and function of PqqF</i> (Wei et al. 2016)	Shows PqqF (M16) performs the essential core PqqA excision and is essential for PQQ synthesis — establishes the two-protease division of labor
28655819	<i>Lanthanides in PQQ-ADHs</i> (Wehrmann et al. 2017)	Characterizes periplasmic PQQ-ADHs PedE/PedH that consume the cofactor PP_0375 helps produce
31427437	<i>A two-component protease in PQQ biosynthesis</i>	Highlights that the identity of the PQQ maturation protease has been historically uncertain and varies between taxa
8606199	<i>pqqE and pqqF in M. extorquens AM1</i>	Historical context: PqqF as a zinc-endopeptidase family member (distinct clan from S9)

Consistency / tension: The literature is internally consistent that PQQ biosynthesis requires a protease for di-amino-acid excision, and that different organisms deploy different peptidases. A key nuance ([PMID: 31427437](#)) is that the maturation protease is a "missing piece" that varies across taxa — in some α -proteobacteria it is a two-component heterodimer, in *Klebsiella*/*Methylobacterium* it is PqqF (M16), and *P. aeruginosa*/*P. putida* additionally carry the S9 *pqqH*/*pqqG*. The evidence assigns the **essential** excision to PqqF in *Pseudomonas* and an **excretion-linked** role to PP_0375/*pqqG* — but does not exclude an accessory maturation role for the S9 enzyme.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of PP_0375 itself.** All functional inference derives from (a) orthology to *P. aeruginosa pqqH* (~53 % identity), (b) genomic operon context, and (c) structural prediction. No *P. putida* PP_0375 knockout, purified enzyme assay, or crystal structure exists in the reviewed literature.
 2. **The molecular substrate is undefined.** Whether PP_0375 cleaves a residual PqqA fragment, a PQQ-maturation intermediate, or an unrelated peptide coupled to export is not established. The "excretion" phenotype of *pqqH* could reflect a direct role in export machinery processing rather than PQQ chemistry.
 3. **The AlphaFold-based active-site and oligopeptidase-selectivity claims are predictions,** not experimental structures — high-confidence (pLDDT 93.8) but unvalidated by crystallography or mutagenesis.
 4. **Localization is inferred.** A cytoplasmic role during maturation is the parsimonious interpretation, but the excretion phenotype raises the possibility of association with the inner membrane or export apparatus; no experimental subcellular-localization data were found.
 5. **Operon transcription nuance:** the KT2440 cluster produces "at least two independent transcripts," so *pqqG* expression may be regulated separately from the core *pqqABCDE*, but this was not investigated in detail here.
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Proposed Follow-up Experiments / Actions

1. **Targeted gene deletion (Δ PP_0375) in *P. putida* KT2440,** measuring (i) intracellular vs. extracellular PQQ (LC-MS or the classic *E. coli* GDH-reconstitution bioassay), (ii) glucose→gluconate flux, and (iii) periplasmic Gcd/PedE-PedH activity. This directly tests whether the *pqqH* "excretion" phenotype transfers to PP_0375.
2. **Recombinant expression and enzymology.** Purify PP_0375 and assay peptidase activity against candidate substrates: synthetic PqqA-derived peptides, cross-linked Glu–Tyr intermediates, and generic oligopeptide libraries (e.g., internally quenched fluorogenic peptides) to define the length and sequence specificity predicted by the gated β -propeller.

3. **Active-site mutagenesis.** Ser469Ala (and His582Ala) catalytic-dead variants to confirm the triad is required for the physiological phenotype, decoupling catalytic vs. possible scaffolding roles.
 4. **Structural validation.** Solve a crystal or cryo-EM structure of PP_0375 to confirm the β -propeller / α - β -hydrolase two-domain arrangement and the buried active site, and to visualize the substrate-access channel.
 5. **Interaction / localization mapping.** Co-purification or BioID with operon partners (PqqF, PqqE, PqqB/C) and with export components; cell-fractionation to test cytoplasmic vs. membrane-associated localization.
 6. **Comparative genomics of the two-protease pattern.** Systematically map which PQQ-producing genomes carry PqqF, *pqqG/pqqH*, both, or the α -proteobacterial two-component protease, to clarify the evolutionary logic of redundancy and the specific selective role of the S9 enzyme.
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Report generated from a 5-iteration autonomous investigation: 6 confirmed findings, 15 papers reviewed. All functional claims are supported by the cited literature; claims resting on structure prediction or orthology are flagged as such.
